

Question	Marking Guidance	Mark	Comments
05.1	Row 2;	1	
05.2	D – Granum/grana/thylakoid(s); E – starch/lipid;	2	Accept oil for E
05.3	1. Light has long(er) wavelength; 2. (So) low(er) resolution;	2	Accept converse in context of electron microscope 1. Ignore: optical microscope has long(er) wavelength. 2. Accept poor resolution 2. Ignore: weaker resolution 2. Ignore references to magnification 2. Accept correct references to values for resolution. E.g optical 0.2µm – 0.3 µm
05.4	(70S) Ribosome;	1	Reject: (80S) Ribosome
05.5	Correct answer of 7455 = 2 marks;; Accept for 1 mark answers in range: 7717.5 to 7718 (44.1% of 17500) If incorrect answer, accept for 1 mark working shows an attempt to subtract 262.5	2	

05.6	1. (Ice) cold to prevent/reduce enzyme activity; 2. Buffered to prevent denaturing of enzyme/protein; 3. Same water potential/Ψ to prevent lysis/bursting (of organelle);	3	For 1, 2 and 3 reject context of cell 2. Accept description of buffer. 2. Accept: prevent change of tertiary structure. 3. Accept: isotonic for same water potential. 3. Reject: references to turgor or plasmolysis or crenation.
TOTAL		11	